

Linear Models Lecture 18: Random Effects

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[illegible]

Motivation for Random Effects/ Partial Pooling

- Given a model $y_{it} = \beta_0 + \beta_1 x_{it} + \alpha_i + e_{it}$, assume that
 - 1 Strict exogeneity holds: $E[e_i | \alpha_i, \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{iT}] = 0 \quad \forall i$
 - 2 The unobserved features are uncorrelated with $\mathbf{X}$. $E[\alpha_i | \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{iT}] = \alpha_0 \quad \forall i$.
- Pooled OLS is consistent under 1), but we can do more (derive a more efficient estimator) with assumption 2).
- Under 2) we can use the group structure to
 - rebalance the between and within estimator, dragging estimates toward group means,
 - use information about variables that do not vary within unit,
 - make predictions about unobserved groups.
- This comes at the cost of introducing bias if 2) is false, ie, we failed to control for some time-invariant characteristic.

Aitken (1935) Generalized Least Squares

- We will use group structure to model dependence between observations.

$$\text{var}[\mathbf{e}|\mathbf{X}] = \Sigma\sigma^2$$

- If we know Σ , we can do no better than to premultiply our linear model with $\Sigma^{-1/2}$,

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$$

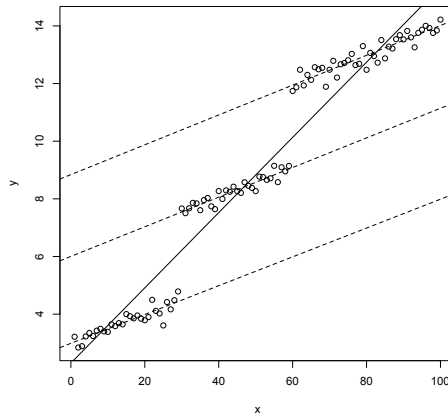
$$\Sigma^{-1/2}\mathbf{Y} = \Sigma^{-1/2}\mathbf{X}\beta + \Sigma^{-1/2}\mathbf{e}$$

$$\tilde{\mathbf{Y}} = \tilde{\mathbf{X}}\beta + \tilde{\mathbf{e}}$$

$$\begin{aligned}\tilde{\beta}_{GLS} &= (\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\tilde{\mathbf{Y}} \\ &= ((\Sigma^{-1/2}\mathbf{X})'(\Sigma^{-1/2}\mathbf{X}))^{-1}(\Sigma^{-1/2}\mathbf{X})'(\Sigma^{-1/2}\mathbf{Y}) \\ &= (\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{Y}\end{aligned}$$

Here we have $E[\tilde{\beta}_{GLS}] = \beta$ and $\text{var}(\tilde{\beta}_{GLS}) = \sigma^2(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}$

Random Effects Assumption



Random Effects Assumption

- Define a composite error term $\nu_{it} = \alpha_i + e_{it}$.

$$y_{it} = \beta' \mathbf{x}_{it} + \alpha_i + e_{it} = \beta' \mathbf{x}_{it} + \nu_{it}$$

$$\begin{aligned} \text{Var}(\nu_{it}) &= E[(\alpha_i + e_{it})^2] = E[\alpha_i^2 + 2\alpha_i e_{it} + e_{it}^2] = \sigma_\alpha^2 + \sigma_e^2 \\ \text{Cov}(\nu_{it}\nu_{is}) &= E[\nu_{it}\nu_{is}] = E(\alpha_i + e_{it})(\alpha_i + e_{is}) \\ &= E(\alpha_i^2 + \alpha_i e_{it} + \alpha_i e_{is} + e_{it}e_{is}) = E(\alpha_i^2) = \sigma_\alpha^2 \quad \forall t \neq s \end{aligned}$$

- Recall correlation is $\rho = \text{cov}(x, y) / \sqrt{\text{var}(x)\text{var}(y)}$

$$\rho_{\nu_{it}\nu_{is}} = \frac{\sigma_\alpha^2}{\sqrt{(\sigma_\alpha^2 + \sigma_e^2)(\sigma_\alpha^2 + \sigma_e^2)}} = \frac{\sigma_\alpha^2}{(\sigma_\alpha^2 + \sigma_e^2)}$$

Building Blocks

Suppose that $\boldsymbol{\nu}_i = \begin{pmatrix} \nu_{i1} \\ \nu_{i2} \\ \vdots \\ \nu_{iT} \end{pmatrix}$ for a generic individual i .

$$\begin{aligned} E(\boldsymbol{\nu}_i \boldsymbol{\nu}_i') &= \begin{pmatrix} \sigma_\alpha^2 + \sigma_e^2 & \sigma_\alpha^2 & \dots & \sigma_\alpha^2 \\ \sigma_\alpha^2 & \sigma_\alpha^2 + \sigma_e^2 & \dots & \sigma_\alpha^2 \\ \dots & \dots & \dots & \dots \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \dots & \sigma_\alpha^2 + \sigma_e^2 \end{pmatrix} \\ &= \sigma_e^2 \mathbf{I} + \sigma_\alpha^2 \mathbf{ii}' \\ &= \boldsymbol{\Omega} \end{aligned}$$

Stacking people

Stacking individuals $\nu = \begin{pmatrix} \nu_1 \\ \nu_2 \\ \vdots \\ \nu_t \end{pmatrix}$:

$$E[\nu\nu']_{(NT \times NT)} = \begin{pmatrix} \Omega & 0 & 0 & \dots \\ 0 & \Omega & 0 & \\ 0 & 0 & \Omega & \\ \dots & & & \end{pmatrix} = \Omega \otimes I$$

Theoretical GLS: Random Effects Estimation

Using the fact that $\text{var}(\nu) = \Omega \otimes I$, we can apply GLS

$$\hat{\beta}_{RE} = (\mathbf{X}'(\Omega \otimes I)^{-1}\mathbf{X})^{-1}\mathbf{X}'(\Omega \otimes I)^{-1}\mathbf{y}$$

But we need estimates of σ_{ν}^2 , σ_e^2 and σ_{α}^2

$$\sigma_{\nu}^2 = \sigma_e^2 + \sigma_{\alpha}^2$$

FGLS for Random Effects

- Run OLS on pooled data and estimate residuals: $\hat{v}$

$$\widehat{\sigma_v^2} = \frac{\sum_{i=1}^N \sum_{t=1}^T \hat{v}_{it}^2}{NT - k}$$

- We still need estimate of either σ_α^2 or σ_e^2 .
- Wooldridge approach: take an average of observed covariances:

$$E \left(\frac{\sum_{s=1}^{T-1} \sum_{t=s+1}^T \hat{v}_{is} \hat{v}_{it}}{N \frac{(T-1)T}{2} - k} \right) = \sigma_\alpha^2$$

$$\widehat{\sigma_v^2} - \widehat{\sigma_\alpha^2} = \widehat{\sigma_e^2}$$

Partial Pooling

- Random effects models are partial pooling:

$$\mathbf{b}_{RE} = \lambda \mathbf{b}_W + (\mathbf{I} - \lambda) \mathbf{b}_B$$

$$\lambda = (\mathbf{S}_{xx}^W + (1 - \theta^2) \mathbf{S}_{xx}^B)^{-1} \mathbf{S}_{xx}^W$$

- Here θ is between 0 and 1. Under random effects:

$$\theta = 1 - \sqrt{\frac{\sigma_e^2}{\sigma_e^2 + T\sigma_\alpha^2}}$$

$$\theta = 1 - \sqrt{\frac{1}{1 + T(\sigma_\alpha^2/\sigma_e^2)}}$$

Comparing Fixed vs Random Effects

■ Random effects

- Is efficient.
- Allows analysis of time-constant variables.
- Allows extrapolation to unobserved groups.
- Assumes $\mathbf{X}$ is independent of α
- `plm(y ~ x, data = dta, index = c("id", "year"), model = "random")`

■ Fixed effects

- Washes out time-constant variables.
- Uses many n degrees of freedom.
- Allows any sort of relationship between $\mathbf{X}$ and α
- `plm(y ~ x, data = dta, index = c("id", "year"), model = "within")`

Hausman (1978) Specification Test

- **Idea:** compare FE and RE estimates of β .
 - Under H_0 (RE valid): both consistent, RE more efficient $\Rightarrow$ difference is small.
 - Under H_1 (RE invalid): RE inconsistent, FE still consistent $\Rightarrow$ systematic difference.
- **Test statistic**

$$H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' [\widehat{\text{Var}}(\hat{\beta}_{FE}) - \widehat{\text{Var}}(\hat{\beta}_{RE})]^{-1} (\hat{\beta}_{FE} - \hat{\beta}_{RE}) \xrightarrow{d} \chi_K^2$$

where $K = \dim(\beta)$. Reject H_0 when H is large.

- **GMM connection:** This is a special case of the GMM endogeneity test (Hansen, Thm 13.16, Lecture 16). The GMM version replaces the classical variance difference with a robust sandwich estimator, making it valid under heteroskedasticity.

Hausman Test in Practice

```
fe_mod <- plm(log(gsp) ~ log(pcap) + log(hwy) + log(util),  
              data = pdata, model = "within")  
re_mod <- plm(log(gsp) ~ log(pcap) + log(hwy) + log(util),  
              data = pdata, model = "random")  
  
phtest(fe_mod, re_mod) # classical Hausman chi-squared
```

- **Caveat:** This is a *specification test*, not a model-selection tool.
- Using it to choose FE vs. RE makes your estimator a *pretest estimator* $\Rightarrow$ biases subsequent inference.
- Better alternatives when unsure:
 - Use FE (conservative).
 - Use **Correlated Random Effects** (Mundlak), which nests both and supplies a robust version of the same test.

Time-Invariant Regressors: the Identification Problem

Panel baseline

$$y_{it} = \mathbf{x}_{it}'\beta + \mathbf{z}_i'\gamma + \alpha_i + e_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

- $\mathbf{x}_{it}$: time-varying covariates
- $\mathbf{z}_i$: *time-invariant* covariates
- α_i : unit fixed effect, unobserved and potentially *correlated* with $\mathbf{x}_{it}, \mathbf{z}_i$
- The **within transformation** eliminates α_i but also wipes out $\mathbf{z}_i$ (no variation over t) $\Rightarrow$ standard FE cannot identify coefficients on $\mathbf{z}_i$.
- E.g. we want to know the coefficient on colonial legacy, ethnic fractionalization, or distance to Russia.

Model-Based Remedies for z_i

■ Correlated Random Effects (Mundlak, 1978)

- Model the dependence of α_i on the *time-varying* regressors: $\alpha_i = \delta_0 + \bar{\mathbf{x}}_i' \boldsymbol{\delta} + \zeta_i$, $\bar{\mathbf{x}}_i \equiv T^{-1} \sum_t \mathbf{x}_{it}$.
- Conditional on $\bar{\mathbf{x}}_i$, treat ζ_i as *random*: $\mathbb{E}[\zeta_i \mid \mathbf{x}_{it}, \mathbf{z}_i] = 0$.
- Estimation via GLS delivers FE-consistent $\hat{\beta}$ and identifies γ on $\mathbf{z}_i$.

■ Hausman–Taylor (1981)

- Partition regressors $\mathbf{x}_{it} = (\mathbf{x}_{1it}, \mathbf{x}_{2it})$, $\mathbf{z}_i = (\mathbf{z}_{1i}, \mathbf{z}_{2i})$ such that $\mathbf{x}_{1it}, \mathbf{z}_{1i}$ are exogenous, $\mathbf{x}_{2it}, \mathbf{z}_{2i}$ possibly endogenous.
- Use $\mathbf{x}_{1it}$ (within variation) and $\mathbf{z}_{1i}$ (between variation) as instruments for $\mathbf{x}_{2it}, \mathbf{z}_{2i}$.
- Two-step IV–GLS attains efficiency under random-effects covariance.

Correlated Random Effects (Mundlak)

■ Structural restriction

$$\alpha_i = \delta_0 + \bar{\mathbf{x}}_i' \boldsymbol{\delta} + \zeta_i, \quad \mathbb{E}[\zeta_i \mid \mathbf{x}_{it}, \mathbf{z}_i] = 0$$

where $\bar{\mathbf{x}}_i = T^{-1} \sum_t \mathbf{x}_{it}$. All correlation between α_i and the time-varying covariates is absorbed by $\bar{\mathbf{x}}_i' \boldsymbol{\delta}$.

■ Substituting into the panel model:

$$y_{it} = \mathbf{x}_{it}' \boldsymbol{\beta} + \mathbf{z}_i' \boldsymbol{\gamma} + \delta_0 + \bar{\mathbf{x}}_i' \boldsymbol{\delta} + \underbrace{\zeta_i + e_{it}}_{\text{composite error}}$$

■ Apply GLS (random-effects weighting) to the augmented model.

- $\hat{\boldsymbol{\beta}}$ identical to FE (we will see why via FWL).
- $\hat{\boldsymbol{\gamma}}$ on $\mathbf{z}_i$ is now identified.
- Testable implication: $H_0: \boldsymbol{\delta} = \mathbf{0}$ reduces CRE to conventional RE.

Why $\hat{\beta}_{CRE} = \hat{\beta}_{FE}$: FWL Intuition

- The CRE regression includes $\mathbf{x}_{it}$, $\bar{\mathbf{x}}_i$, $\mathbf{z}_i$, and a constant.
- By the **Frisch–Waugh–Lovell theorem**, the coefficient on $\mathbf{x}_{it}$ is obtained by first *partialing out* $\bar{\mathbf{x}}_i$ (and $\mathbf{z}_i$, constant) from both y_{it} and $\mathbf{x}_{it}$.
- Partialing $\bar{\mathbf{x}}_i$ from $\mathbf{x}_{it}$:

$$\mathbf{x}_{it} - \bar{\mathbf{x}}_i = \text{within-demeaned data}$$

- So $\hat{\beta}_{CRE}$ uses *only within-unit variation* — exactly the same variation that FE uses.
- **Bonus:** CRE also identifies γ (the coefficient on $\mathbf{z}_i$) because ζ_i is uncorrelated with all regressors by assumption.

The Mundlak Test = Hausman Test

- From the CRE regression, test

$$H_0: \delta = \mathbf{0} \quad (\text{unit means have no additional explanatory power})$$

- Under H_0 : α_i uncorrelated with $\mathbf{x}_{it} \Rightarrow$ standard RE is valid.
- Under H_1 : RE is inconsistent, but CRE (= FE for β) remains consistent.
- A standard F -test (or Wald test) on $\delta = \mathbf{0}$ is **numerically equivalent** to the classical Hausman χ^2 .
- **Advantage:** the CRE version
 - is easy to compute (just an F -test on extra regressors),
 - works with *robust* / *clustered* standard errors (the classical Hausman test does not).

R implementation: Mundlak CRE

```
library(plm)                                # panel-data toolkit
data("Produc", package = "Ecdat")# US state production data

dffrost<-data.frame( state = str_replace(toupper(rownames(state.x77)), "\\s", "_"),
frost = state.x77[, "Frost"])%>%mutate(state=ifelse(state=="TENNESSEE", "TENNESSE", state))

Producweather<-Produc%>%left_join(dffrost)
pdata <- pdata.frame(Producweather, index = c("state", "year"))

pdata$pcap_bar <- Between(log(pdata$pcap))    # == ave(pcap, state)
pdata$hwy_bar <- Between(log(pdata$hwy))
pdata$util_bar<- Between(log(pdata$util))

cre_mod <- plm(log(gsp) ~ log(pcap)+log(hwy)+log(util)+pcap_bar+hwy_bar+util_bar+frost,
               data    = pdata,
               model    = "random")           # GLS w/ error-components

fe_mod <- plm(log(gsp) ~ log(pcap)+ log(hwy)+log(util),
              data    = pdata,
```

Regression results

	Model 1	Model 2
Intercept	0.14 (0.65)	
<i>Time-varying covariates</i>		
log(pcap)	2.26*** (0.15)	2.26*** (0.15)
log(hwy)	-0.81*** (0.11)	-0.81*** (0.11)
log(util)	-0.47*** (0.08)	-0.47*** (0.08)
<i>Means</i>		
pcap_bar	-1.19 (0.81)	
hwy_bar	0.87 (0.45)	
util_bar	0.41 (0.40)	
<i>Time invariant</i>		
frost	-0.00 (0.00)	
σ_{idios}	0.08	
σ_{id}	0.18	
R^2	0.81	0.71
Adj. R^2	0.81	0.69
Observations	816	816
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$		

Two-Way Mundlak (Wooldridge, 2021)

- With unit *and* time fixed effects: $y_{it} = \mathbf{x}'_{it}\beta + \alpha_i + \lambda_t + e_{it}$
- **Two-way Mundlak:** include both *unit means* $\bar{\mathbf{x}}_i$ and *time means* $\bar{\mathbf{x}}_t$ as additional regressors in a pooled or RE regression.
- Delivers TWFE-equivalent $\hat{\beta}$ while allowing
 - time-invariant regressors ($\mathbf{z}_i$) and
 - unit-invariant regressors ($\mathbf{w}_t$).
- **Key application:** staggered difference-in-differences with heterogeneous treatment effects.
 - Standard TWFE can assign *negative weights* to some treatment effects.
 - Two-way Mundlak avoids this problem by explicitly modeling the dependence structure.

Robust Standard errors

- Newey-West, like White with a specified lag, adjusts for autocorrelation and heteroskedasticity. `vcovNW`
- Beck-Katz, use large- T asymptotics to correct for cross-sectional dependence: good for if T/N is not too small. `vcovBK`
- Driscoll-Kraay applies Newey-West to cross-sectional dependence: good if T/N is small. `vcovSCC`

The Dynamic Panel Problem

- Many panel models include a lagged dependent variable:

$$Y_{it} = \rho Y_{it-1} + \mathbf{X}_{it}'\beta + \alpha_i + e_{it}$$

- **Problem:** the within transformation creates correlation between the transformed lagged DV and the transformed error (Nickell, 1981):

$$\tilde{Y}_{it-1} = (Y_{it-1} - \bar{Y}_i) \quad \text{is correlated with} \quad \tilde{e}_{it} = (e_{it} - \bar{e}_i)$$

because $\bar{Y}_i$ contains Y_{it} which depends on e_{it} , and $\bar{e}_i$ contains e_{it-1} which affects Y_{it-1} .

- The bias is $O(1/T)$: severe when T is small (typical in micro panels), vanishes as $T \rightarrow \infty$.
- **FE is inconsistent** for fixed T with a lagged dependent variable.

Arellano-Bond: First Differencing + Instruments

- **Step 1:** First-difference to eliminate α_i :

$$\Delta Y_{it} = \rho \Delta Y_{it-1} + \Delta \mathbf{X}_{it}' \boldsymbol{\beta} + \Delta e_{it}$$

- **Step 2:** ΔY_{it-1} is correlated with Δe_{it} (since Y_{it-1} appears in both).
Use **lagged levels** as instruments: Y_{is} for $s \leq t-2$.

- **Moment conditions:**

$$E[Y_{is} \cdot \Delta e_{it}] = 0 \quad \text{for } s \leq t-2$$

- At $t = 3$: one instrument (Y_{i1}). At $t = 4$: two (Y_{i1}, Y_{i2}). At $t = T$: $T-2$ instruments.
- Total moment conditions: $\frac{(T-1)(T-2)}{2}$ — the system is **overidentified** $\Rightarrow$ GMM.

Arellano-Bond as GMM

- This is exactly the GMM framework from Lectures 15–16:
 - 1 **Step 1:** Estimate with a preliminary weight matrix (e.g., $\mathbf{W} = (\mathbf{Z}'\mathbf{H}\mathbf{Z})^{-1}$ where $\mathbf{H}$ is a first-difference covariance structure).
 - 2 **Step 2:** Re-estimate using the optimal weight matrix from step 1 residuals (two-step GMM).
- **J-test** (Hansen/Sargan): test the $\frac{(T-1)(T-2)}{2} - k$ overidentifying restrictions.
 - df = moments – parameters
 - Rejection $\Rightarrow$ serial correlation in e_{it} or invalid instruments.
- **Key diagnostic:** AR(2) test on differenced residuals.
 - AR(1) in Δe_{it} is expected by construction.
 - AR(2) in Δe_{it} implies AR(1) in $e_{it} \Rightarrow$ instruments invalid.

Extensions and Practical Guidance

- **System GMM** (Blundell-Bond, 1998): augments the differenced equations with *level equations* instrumented by lagged differences.
 - Additional moments: $E[\Delta Y_{is} \cdot (\alpha_i + e_{it})] = 0$ for $s = t - 1$.
 - Especially useful when the series is persistent (ρ near 1), where Arellano-Bond instruments are weak.
- **Practical rules:**
 - Designed for “large N , small T ” panels.
 - Too many instruments $\Rightarrow$ overfitting. Collapse or limit lag depth.
 - Always report: (1) J -test p -value, (2) AR(2) test, (3) number of instruments vs. groups.
- **Software:** `plm::pgmm()` in R; `xtabond2` in Stata.
 - `pgmm(y ~ lag(y,1) + x | lag(y, 2:99), data=pdata, effect="twoways")`

Application: Does Income Cause Democracy?

- **Acemoglu, Johnson, Robinson & Yared (2008):** Classic question — does economic development promote democratization?
- **Model:** $\text{Democracy}_{it} = \rho \text{Democracy}_{it-1} + \beta \text{Income}_{it-1} + \alpha_i + \lambda_t + e_{it}$
- **The problem with FE:** Democracy is highly persistent ($\rho \approx 0.7$).
 - Within-transformation induces Nickell bias in $\hat{\rho}$ (biased toward zero).
 - This bias contaminates $\hat{\beta}$ as well.
 - With $T \approx 10$ (five-year panels), the bias is substantial.
- **Arellano-Bond solution:** First-difference to eliminate α_i , then instrument $\Delta \text{Democracy}_{it-1}$ with lagged levels.
- **Finding:** Once dynamic panel bias is corrected, the income effect on democracy becomes insignificant — country fixed effects explain most of the cross-national correlation.

Arellano-Bond in R: Tax Progressivity Example

```
library(plm); library(haven)
dta <- read_dta("progressTax.dta")
pdta <- pdata.frame(subset(dta, year >= 1850),
                    index = c("ccode", "year"))

# Arellano-Bond: two-step GMM
ab_mod <- pgmm(
  topratep ~ lag(topratep, 1) + himobpopyear2p
    | lag(topratep, 2:99),
  data = pdta,
  effect = "twoways",      # unit + time effects
  model = "twosteps")     # two-step GMM
summary(ab_mod)

# Key diagnostics
summary(ab_mod)$sargan    # J-test (overid)
# Check AR(2) test: p > 0.05 validates
# the moment conditions
```

Reading Arellano-Bond Output

```
summary(ab_mod)
# Coefficients:
#               Estimate Std. Error z-value Pr(>|z|)
# lag(topratep,1)    0.xxx      0.xxx    x.xx   0.xxx
# himobpopyear2p     0.xxx      0.xxx    x.xx   0.xxx
#
# Sargan test: chisq(df) = xx, p-value = 0.xxx
# (large p -> instruments valid)
# Autocorrelation test (2): z = x.xx, p = 0.xxx
# (large p -> no AR(2) in levels)
```

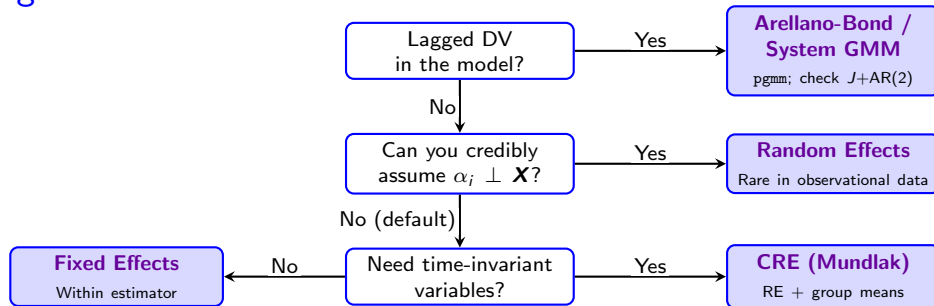
Checklist for reporting:

- 1 $\hat{\rho}$: persistence of lagged DV (compare to FE estimate)
- 2 Sargan/Hansen J -test p -value > 0.05
- 3 AR(2) test p -value > 0.05
- 4 Number of instruments $<$ number of groups

Critical Assessment: When Lagged-Instrument GMM Works

- **What the exclusion restriction really requires.** Lagged levels Y_{is} ($s \leq t - 2$) are valid instruments for ΔY_{it-1} **only if e_{it} is serially uncorrelated**. This is a substantive assumption about the structural error — not a regularity condition — and it is testable (AR(2) test on differenced residuals).
- **Where the technique struggles in applied work.**
 - **Persistent series** ($\rho \rightarrow 1$): Arellano-Bond instruments become weak; Blundell-Bond System GMM was developed for this case but inherits its own validity concerns.
 - **Short panels with many lags:** the instrument count grows quadratically in T . Roodman (2009) shows the Sargan test loses power as instruments proliferate — passing it stops being informative. “Collapse lags” is a band-aid, not a fix.
 - **Plausible serial correlation in e_{it} :** most political-economy outcomes (vote shares, conflict, regime quality) have persistent unobserved drivers. When AR(2) rejects, the moment conditions fail by your own diagnostic.
- **Why it's less common today.** Applied work has shifted toward design-based identification — DiD with staggered adoption, RD, shift-share IV with explicit exposure

Choosing a Panel Estimator



The choice of estimator should follow from your research design, not from a post-hoc test.

Why Not Let the Hausman Test Decide?

- A common workflow: estimate FE and RE, run the Hausman test, pick the “winner.”
- This is **pre-test estimation** — your final estimator is selected by a preliminary test.
- **Problem:** the sampling distribution of the reported estimate is a *mixture* of the FE and RE distributions, weighted by the probability the test selects each one.
- Standard errors and confidence intervals ignore the selection step $\Rightarrow$ **undercoverage**.
- **Better approach:** let your **research design** determine the estimator:
 - Is selection into units plausibly correlated with covariates? (Almost always yes with observational country-, state-, or individual-panels.) $\Rightarrow$ FE or CRE.
 - Use the Hausman test as a *diagnostic*, not a decision rule. Report it alongside your preferred specification.

When Is Random Effects Reasonable?

RE assumes $\alpha_i \perp \mathbf{X}_{it}$: unit heterogeneity is unrelated to covariates.

Plausible settings in political science:

- 1 **Randomized/quasi-random assignment across units** — e.g., audit experiments in randomly sampled municipalities. Treatment is exogenous.
- 2 **Surveys with cluster sampling** — individuals within randomly selected clusters. The cluster effect is a nuisance, not a confounder.
- 3 **Meta-analysis** — studies are “units”; heterogeneity is modeled as random for population-average inference.

When to be skeptical (most observational panels):

- Countries self-select into trade agreements, wars, institutions.
- States adopt policies endogenously.
- Individuals sort into groups based on characteristics that also affect Y .

Default: assume α_i is correlated with $\mathbf{X}$ and use FE or CRE unless you can argue otherwise.

Discussion and Course Integration

- Panel estimators fit into the **unified GMM framework** from Lectures 15–16:
 - FE: within-group moment conditions (just-identified).
 - RE: adds between-group moments via GLS weighting (efficiency gain under H_0).
 - CRE (Mundlak): augmented RE that nests FE; Mundlak test = Hausman test.
 - Arellano-Bond: lagged levels as instruments in first-differenced equations (overidentified $\Rightarrow J$ -test).
- You can use panel structure to improve efficiency (not discarding cross-unit variation) at the cost of bias if assumptions fail.
- Many studies focus on slow-moving processes that cannot be estimated with pure fixed effects.
- Dynamic panels require GMM-based approaches; the J -test provides a specification check that was not available with FE alone.